

# An Interactive Framework for Named Entity-Based Reproducibility Analysis of Bioinformatics Workflows

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**Abstract.** Computational workflows are widely used in bioinformatics to automate complex data analysis. Ideally, it should be possible to match tools reported in a publication to the corresponding implementation in executable workflow code; discrepancies between the two may indicate potential reproducibility and reusability issues. Developing on CoPaLink, a research pipeline that leverages Natural Language Processing (NLP) methods to identify and match bioinformatics tools reported between publications and workflow code. This work extends CoPaLink through an accessible web application for interactive analysis and comparison of tools reported in article and workflow code. Overall, the interface integrates tool extraction, comparison, and contextual inspection, providing a practical means of investigating consistency between reported and implemented computational workflows in support of reproducibility analysis.

**Keywords:** Named Entity Recognition · Bioinformatics Workflows · Reproducibility

## 1 Context and Research Problem

Reproducing and reusing computational analyses requires establishing a clear correspondence between what is described in scientific publications and what is actually implemented in code. Verifying this consistency remains a challenging and largely manual process. In bioinformatics, analyses often rely on complex computational workflows [1] that integrate multiple tools. These workflows may be described in articles and shared through public repositories. Ensuring reproducibility, therefore, requires reconciling the tools described in publications with their actual use in workflow code. This comparison is further complicated by differences in representation: publications describe methods through unstructured natural language, whereas workflow repositories encode computational steps in code. Tool names may also differ across these sources: for example, *Krona* may be mentioned by name in a publication while the corresponding workflow invokes the executable `ktImportTaxonomy`.

This project extended CoPaLink [2], by developing an interactive web interface that makes it accessible to users and facilitates the exploration of identified links between article text and code. The interface enables researchers to inspect extracted tools, their correspondences, and discrepancies within the original publication and workflow contexts, thereby supporting the analysis of computational reproducibility in bioinformatics.

## 2 Proposed Approach

The extraction pipeline considers two complementary sources of information: scientific publications and workflow repositories. Scientific article text and workflow code are analyzed for extracting tool mentions using specific Name Entity Recognition (NER) models based on encoder language models, SciBERT [3] and CodeBERT [4] respectively. The extracted mentions are subsequently passed to Sentence-BERT [5] for their matching, and the resulting annotations and tool correspondences are returned to the interface for visualization and interactive inspection. The overall architecture with extraction and matching process employed in this work is illustrated in Fig. 1(a).

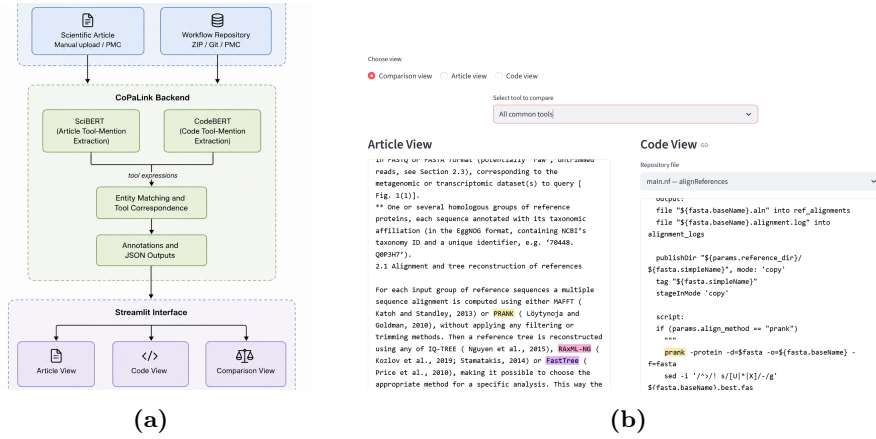
The application supports both manual input and automated retrieval through PubMed Central (PMC), a full-text archive of biomedical and life-sciences literature. Users may provide a scientific article and a workflow repository manually, with workflow code accepted as a ZIP archive or obtained from a Git repository. Alternatively, a PMC identifier can be used to automatically retrieve the corresponding publication and identify associated workflow repositories.

## 3 Implemented Contribution

The main contribution of this internship was the development of a Streamlit-based interactive framework that facilitates the execution of the CoPaLink pipeline and inspecting its results.

For publications and workflow code, extracted tool mentions are highlighted using the positions provided by the pipeline annotations. Users can select tools and navigate between their occurrences while retaining the surrounding article or code context. In the combined analysis, the interface presents tools common to both sources and those occurring exclusively in either source. Many-to-many correspondences produced by the matching pipeline are preserved, allowing relationships between multiple tool occurrences to be inspected rather than reducing them to one-to-one matches. An example of the comparison interface is shown in Fig. 1(b). The interface additionally presents similarity and coverage statistics for analysis. The interface also allows users to download the raw JSON output generated by CoPaLink for further inspection or downstream processing.

One characteristic of the proposed tool is to rely on encoder-based language models, which are lightweight compared to the large language models that are now widely used. As an illustrative benchmark, and without specific optimization, particularly with respect to model loading, five article–workflow pairs were



**Fig. 1.** (a) Architecture of the developed application. (b) Comparison view of bioinformatics tools identified in the scientific article and workflow code.

analyzed concurrently five times on both a CPU and an NVIDIA RTX A6000 GPU. The total execution time was 42.43 s on the CPU and 45.99 s on the GPU. These results indicate that the proposed tool can be effectively deployed on lightweight hardware configurations, including those equipped with a CPU.

## 4 Conclusion

This internship integrated CoPaLink information extraction and entity-matching pipeline into an interactive web application for reproducibility analysis in bioinformatics. The resulting interface enables researchers to execute the pipeline and inspect extracted tools and their correspondences within scientific publications and executable workflow code, providing a practical environment for investigating consistency between reported and implemented computational workflows.

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